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Prevalence and outcomes of in-hospital cardiac arrest in the intensive care unit: a systematic review and meta-analysis

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Abstract

Background Cardiac arrest in the intensive care unit (ICU-CA) is distinct from other in-hospital cardiac arrests, involving critically ill patients in monitored settings. Its prevalence and outcomes remain unclear. This systematic review and meta-analysis aimed to fill this evidence gap and identify areas for future research to improve outcomes in this patient population.

Methods We searched MEDLINE, Embase and Scopus databases from inception until 26 November 2025 for studies reporting on ICU-CA in adults. We performed random effects meta-analyses with the generalised linear mixed model. We used the Joanna Briggs Institute Checklist to assess risk of bias and the GRADE approach to assess the certainty of evidence. The primary outcome was the prevalence of patients with ICU-CA; secondary outcomes included ICU and in-hospital mortality. We performed subgroup analyses based on geographical region (continent), study source (registry vs. non-registry), and COVID-19 status (infected vs. non-infected).

Results We included 35 observational studies including 36 cohorts in the meta-analysis. The pooled proportion of ICU-CA was 3.23% (95% CI: 2.08–4.98, moderate certainty). The proportion of ICU-CA was significantly higher ($p < 0.001$) in patients with COVID-19 (10.6%, 95%-CI: 5.4–19.7) compared with non-COVID-19 cohorts (2.3%, 95%-CI: 1.5–3.5). Pooled ICU mortality was 74.0% (95%-CI: 63.7–82.1, moderate certainty) and in-hospital mortality was 82.0% (95%-CI: 75.9–86.9, high certainty). Meta-regression found that proportion of shockable rhythm was inversely associated with ICU mortality, but not in-hospital mortality.

Conclusion Cardiac arrest in the ICU was associated with poor survival, with a higher prevalence in patients with COVID-19. Future studies should focus on peri-arrest characteristics to guide prognostication and management of individuals who experience ICU-CA.

Keywords Cardiac arrest, Intensive care unit, Cardiopulmonary resuscitation, Prevalence, Survival

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Introduction

In-hospital cardiac arrest (IHCA) is estimated to occur in over 290,000 individuals per year in the United States with approximately 25% of patients surviving to discharge [1, 2]. An epidemiological study on IHCA in Australia and New Zealand hospitals found that 53.4% of IHCAs occurred in monitored areas [3]. While intensive care units (ICUs) are a common location for IHCA [4], accounting for a substantial number of cases, cardiac arrest within the ICU (ICU-CA) remains under-investigated as a distinct clinical entity.

Inherent differences exist between ICU-CA and those that occur outside the ICU, making extrapolation of data to this cohort challenging. The monitored environment of an ICU and higher staff-to-patient ratio usually translates to a more rapid response compared to lower acuity settings; on the other hand, patients in ICU also have greater burden of comorbidities and illness severity. These prognostic factors have been shown to influence outcomes in IHCA, but their impact on outcomes in ICU-CA remains unclear [5, 6].

Previous reviews have attempted to quantify the incidence and outcomes of ICU-CA [7, 8], but they were limited by sample size and did not attempt to analyse subgroups or study-level covariates. Since then, more data within the field have been published, including studies on individuals with COVID-19 and cardiac arrest [9], and studies derived from large-scale registries. As the complications of COVID-19 infection, including cardiac and respiratory injury, became better elucidated [10, 11], more studies on IHCA in the COVID-19 population were conducted. Reports have also suggested more critical illness and organ dysfunction with COVID-19 compared to other pathogens such as influenza [12, 13]. Accordingly, we performed a systematic review and meta-analysis on the prevalence of ICU-CA, and report on relevant post-arrest outcomes, to identify areas for future research to improve outcomes in this patient population.

Methods

We reported this systematic review in line with the PRISMA guidelines (Table S1) [14], prospectively registering the review with PROSPERO (CRD420251064966).

Search strategy and selection criteria

We searched MEDLINE (Ovid), Embase (Ovid) and Scopus databases from inception through 26 November 2025 for literature on ICU-CA without language restrictions, using the key words “intensive care units”, “heart arrest” and “cardiopulmonary resuscitation” (Table S2). We included studies reporting on adults (≥ 18 years) who experienced ICU-CA, or in areas with similar levels of monitoring like critical care units and cardiac/respiratory care units. We excluded articles reporting on cardiac

arrest in other settings (e.g. out-of-hospital cardiac arrest, or IHCA outside or en route to the areas stated above), and studies that did not explicitly report on ICU-CA. Reviews, meta-analyses, editorials, commentaries, conference abstracts and case series were excluded.

To prevent duplication of patient datasets, studies using data from the same registry or originating from the same centre(s) during overlapping time periods were collated. The most appropriate study was selected based on predefined selection criteria which included (1) number of outcomes of interest and (2) sample size, with preference given to studies with greater relevance and size respectively. For our prevalence analysis, we included all patients who experienced an ICU-CA, regardless of whether CPR was attempted to reflect the true prevalence of ICA-CA. For all other outcomes, patients with a ‘do-not-resuscitate’ order were excluded from analysis, to ensure that outcomes were not falsely influenced by patients excluded from resuscitation. While some studies did not provide information on prevalence, they reported on key secondary and tertiary outcomes. We included them in specific pooled analyses and identify them in Table S3.

Data collection and outcomes

Two authors (OL, VM) screened the articles, collected the data, and assessed the risk of bias independently and in duplicate. Conflicts were resolved by a third author (AS). We extracted data using a pre-specified data sheet (Table S4). The primary outcome was the proportion of patients with ICU-CA. We allowed for ICU readmissions with our prevalence analysis. Only the first cardiac arrest episode during an ICU admission was considered. Secondary outcomes included ICU, in-hospital and one-month mortality. Tertiary outcomes included ICU and in-hospital length of stay and functional recovery, with good neurological outcome defined as a modified Rankin Score (mRS) of 0–3 or a Cerebral Performance Category (CPC) score of 1–2 [15]. Secondary and tertiary outcomes excluded re-admissions from the pooled sample. This was in keeping with data reporting amongst included studies. Data were pooled if there were ≥ 3 studies that reported on the outcome of interest.

Risk of bias assessment and certainty of evidence

We assessed risk of bias using the Joanna Briggs Institute (JBI) checklist for prevalence studies [16]. Publication bias was assessed for by visually inspecting funnel plots, Egger’s test and the trim-and-fill procedure. The trim-and-fill method incorporates potentially missing studies due to publication bias, with an underlying assumption that extreme effect sizes are suppressed, and provides a newly adjusted effect estimate [17]. As inter-study heterogeneity between observational studies tends to be

overestimated by I^2 statistics, we assessed heterogeneity using the GRADE approach [18], accounting for both quantitative heterogeneity (I^2 statistics, tau-squared values) and qualitative heterogeneity (distribution of point estimates and overlap of 95%-CIs in forest plots).

Statistical analysis

We utilised R (Version 2025.05.1 + 513) for our analysis and considered a p -value of < 0.05 as statistically significant. We estimated values of the mean and standard deviation as per Wan et al. [19]. We performed a single arm random-effects meta-analysis with the generalised linear mixed model (GLMM), which implicitly uses the logit transformation, and computed the 95% confidence interval (95%-CI) with the Clopper-Pearson method [20]. Binary outcomes were reported as pooled proportions, and continuous outcomes as pooled means, each with their corresponding 95%-CIs. We performed sensitivity analyses excluding studies with a high risk of bias (JBI < 7) and studies with special populations.

We conducted post-hoc subgroup analyses based on geographical region (continent), study source (registry vs. non-registry) and COVID-19 status (infected vs. non-infected). Univariable meta-regression was also performed when at least 6 datapoints were reported for potentially significant study-level covariates. Bubble plots were created to illustrate the relationship between selected variables and outcomes.

Results

Summary of publications

Of 10,391 articles screened, we retrieved 160 articles for full-text screening. Ultimately, 35 observational studies, consisting of 36 cohorts (601,777 patients with ICU-CA) were included in our review (Fig. 1) [21–55]. Studies were conducted across North America [21, 22, 24, 25, 33, 35, 38, 39, 42, 44, 46, 47, 51], Europe [28–31, 34, 36, 41, 45, 48–50, 55], Asia [23, 26, 27, 37, 40, 53, 54], Oceania [32, 52], and South America [43]. Based on the World Bank income classification at time of publication, three of the included studies were conducted in upper-middle-income countries [27, 37, 43]. All other studies were conducted in high-income countries. Eight of the articles were registry-based studies [21, 25, 29, 31, 33, 42, 46, 55]. Special populations included COVID-19 cohorts [21, 24, 28, 35, 49], the very elderly (≥ 90 years) [32, 34], patients with active malignancy [37], and those treated in the neurosurgical ICU [54]. Three studies included ICU readmissions, accounting for an additional 86,917 admissions [29, 31, 55]. The mean age of patients was 65.2 years (95%-CI: 62.0–68.3) and the proportion of males was 63.7% (95%-CI: 61.1–66.2%). The risk of bias assessment and GRADE assessment are provided in Tables S5 and

S6. Apart from four studies [39, 43, 50, 51], all remaining studies were of low risk of bias (JBI > 7).

Prevalence of cardiac arrests

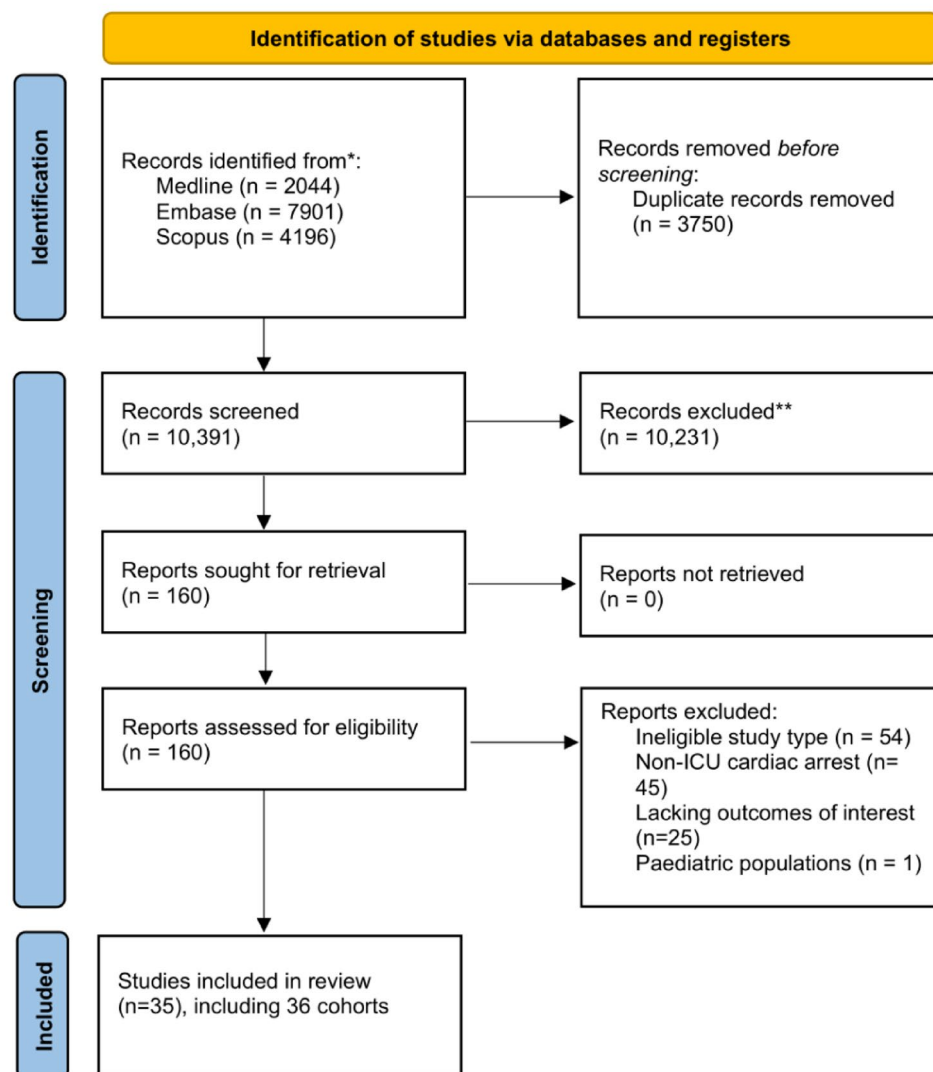
23 studies totalling 5,950,201 patients admitted to the ICU reported on the prevalence of ICU-CA, with a pooled prevalence of 3.23% (95%-CI: 2.08–4.98%, moderate certainty) (Fig. 2). With our subgroup analysis, we found a significantly higher prevalence of ICU-CA ($p < 0.001$) in COVID-19 individuals (10.6%, 95%-CI: 5.4–19.7%, 5 studies) as compared with non-COVID-19 cohorts (2.3%, 95%-CI: 1.5–3.5%, 18 studies) (Fig. 3). There were no differences in prevalence between registry and non-registry studies ($p = 0.42$) or geographical region when comparing between studies in North America, Europe and Asia ($p = 0.52$) (Figures S1–S2, Table S7). With our meta-regression, we found no significant association between prevalence and publication year ($p = 0.32$), SAPS ($p = 0.41$), SOFA ($p = 0.74$), proportion of males ($p = 0.99$) or mean age ($p = 0.06$) (Table S8). The pooled prevalence of ICU-CA after omitting studies with special populations was 1.89% (95%-CI: 1.24–2.88%) (Figure S3). After the addition of 10 studies to account for the effect of missing studies, we found the prevalence of ICU cardiac arrest to be 7.82% (95%-CI: 4.50–13.40%, Egger's test $p = 0.01$) (Figure S4). The four studies with high risk of bias did not report on prevalence of ICU-CA, and a sensitivity analysis excluding these studies was therefore not performed.

ICU mortality

The pooled ICU mortality across 13 studies and 2,724 patients with ICU-CA was 74.0% (95%-CI: 63.7–82.1%, moderate certainty) (Fig. 4A). There were no significant differences in ICU mortality between COVID-19 cohorts ($p = 0.36$) and geographical region ($p = 0.29$) (Table S7). Meta-regression among 13 studies based on proportion of the cohort with shockable rhythms during cardiac arrest found a negative association with ICU mortality (regression coefficient -0.046 , 95%-CI: -0.072 to -0.020 , $p = 0.002$) (Fig. 5). Meta-regression with other study level-covariates including publication year ($p = 0.27$), SAPS ($p = 0.32$), SOFA ($p = 0.22$), proportion of males ($p = 0.41$) and mean age ($p = 0.50$) were not significantly associated with ICU mortality (Table S8).

In-hospital mortality

In-hospital mortality was pooled across 32 studies and 595,318 patients with ICU-CA (82.0%, 95%-CI: 76.2–86.5%, high certainty) (Fig. 4B). There were no differences in in-hospital mortality between COVID-19 ($p = 0.92$), registry-based studies ($p = 0.27$) or geographical region ($p = 0.72$) (Table S7). Meta-regression including 7 studies, and 5,220 patients found a negative association between

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only**Fig. 1** PRISMA Flowchart

SOFA score and in-hospital mortality (regression coefficient -0.481 , 95%-CI: -0.879 to -0.083 , $p = 0.027$) (Figure S5). No association was observed with the other covariates including publication year ($p = 0.13$), SAPS ($p = 0.70$), proportion of males ($p = 0.44$), mean age ($p = 0.24$) and proportion of shockable rhythm ($p = 0.20$) (Table S8). A sensitivity analysis including four studies and 623 patients who experienced cardiac arrest in the coronary care unit, cardiac or cardiothoracic ICU showed an in-hospital mortality of 81.0% (95%-CI: 60.3–92.3%) (Table S7, Figure S6). After excluding four studies with high risk of bias, we found a similar in-hospital mortality rate across 28 studies and 594,944 ICU-CA (81.2%, 95%-CI: 75.0–86.2%) (Table S7, Figure S7).

Other outcomes

One month mortality was 86.7% (95%-CI: 22.4–99.3%, low certainty) when pooled across 4 studies and 489 patients. ICU length of stay was 10.0 days (95%-CI: 5.9–14.1, 11 studies, moderate certainty) and the in-hospital length of stay was 15.8 days (95% CI: 8.9–22.8, 5 studies, moderate certainty). Amongst 515 individuals who survived to hospital discharge, 74.4% (95% CI: 62.2–83.7, moderate certainty) achieved a good neurological outcome. (Figures S8–S11).

Discussion**Key findings**

Our systematic review and meta-analysis, covering nearly 6 million ICU admissions across five continents, found

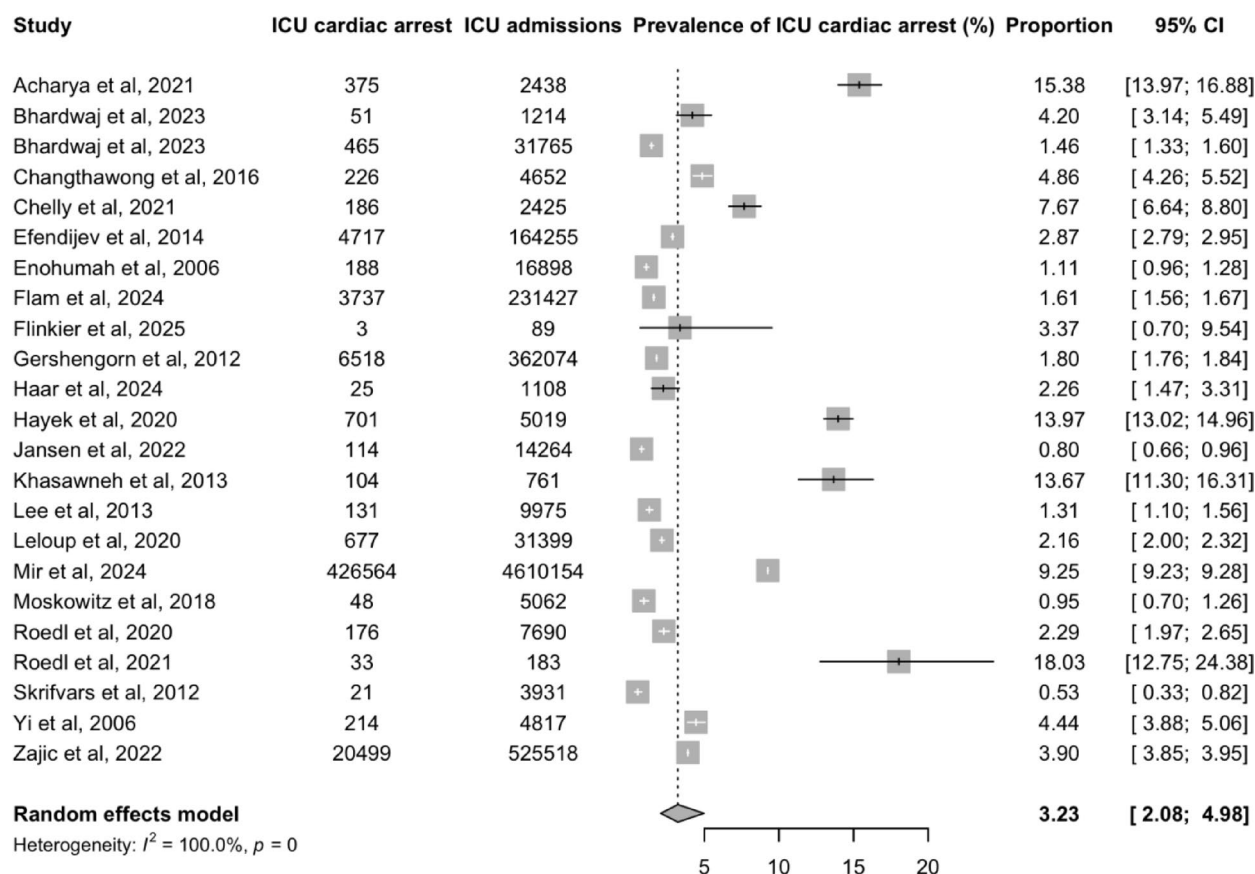


Fig. 2 Forest plot of pooled prevalence of ICU cardiac arrest

that ICU-CA occurred in an estimated 3.2% of admissions to the ICU. Despite happening in a monitored environment, only one in four survived to ICU discharge and about one in five survived their hospital stay, with a shockable ICU-CA rhythm associated with ICU survival. Our findings extend current knowledge on IHCA by estimating the global burden of ICU-CA. Nonetheless, the true prevalence of ICU-CA may be higher than reported given the significant publication bias we had identified in our study, and the corresponding rise in prevalence after adjusting for the effect of missing studies.

COVID-19

Our analysis revealed a significantly higher prevalence of ICU-CA in COVID-19 populations. The largest of these studies (STOP-COVID, 5,000 adults) found that 14.0% suffered a cardiac arrest within 14 days of ICU admission [35]. On multivariable analysis, being admitted to a smaller ICU (< 50 beds), older age, and disease severity were associated with ICU-CA. Critically ill patients with COVID-19 have demonstrated arrhythmogenic manifestations as a disease complication [56]. The profound respiratory failure in these patients, some of whom go on to develop acute respiratory distress syndrome, likely amplifies the risk of developing cardiac arrest [57].

Majority of cardiac arrests (72–95%) amongst included COVID-19 studies were non-shockable (asystole or pulseless electrical activity), supporting a non-cardiac, likely hypoxic aetiology in infected individuals. Although the COVID-19 epidemic has stabilised, the elevated arrest burden may reflect resource strain and altered triage practices during pandemic surges [58, 59], with implications for the preparedness and provision of ICU services in future outbreaks of infectious diseases.

Survival and functional outcomes

Following an ICU-CA, we found that a quarter of individuals survived their ICU admission, and less than one-fifth survived to hospital discharge. This discrepancy highlights a post ICU-CA ‘arrest gap’ in a subset of patients in whom there is a sustained, high risk of death despite initial stabilisation efforts in the ICU. Our findings underline the poor prognosis that ICU-CA portends. Despite this, some registry datasets appear to suggest improved survival over the years, both in the short term [25] and in the longer-term [31]. Demonstrating these temporal trends may be useful but are only possible with robust datasets. It is therefore unsurprising that a paucity of global data exists on the topic, with our included studies illustrating this geographical disparity. Nevertheless,

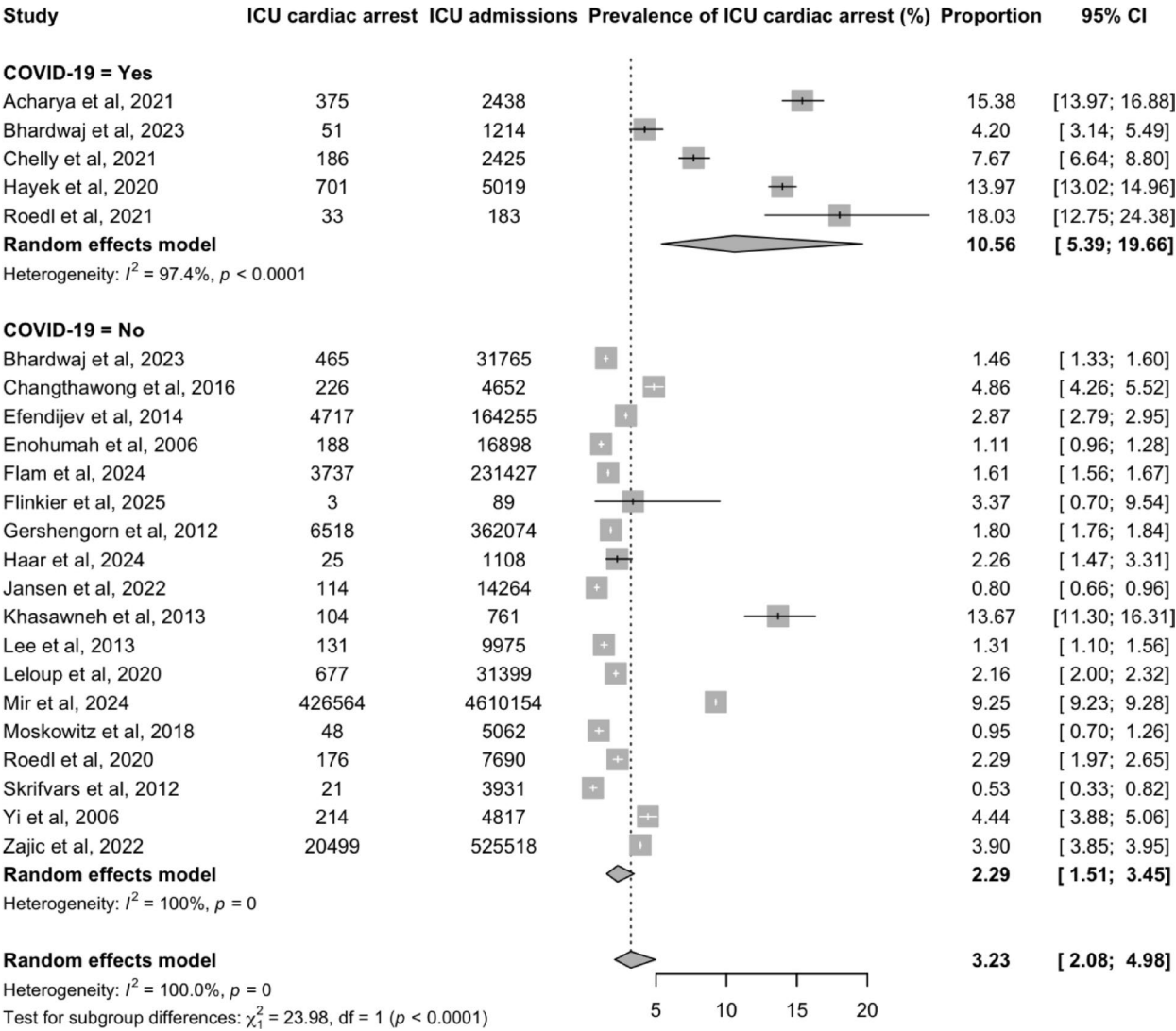


Fig. 3 Forest plot of pooled prevalence of ICU cardiac arrest in COVID-19 and non-COVID-19 cohort

a strategy where institutions collect data tailored to their priorities and available resources can help to reduce the data burden. Having a registry has been incorporated into one of the ten steps towards improving IHCA quality and outcomes [60].

Amongst individuals who survived to hospital discharge, we found that 74.4% had a favourable neurological outcome. This is consistent with prior studies examining functional outcomes following ward-based IHCA [61, 62]. While our study reports novel data on neurological recovery in ICU-CA, data was pooled across a relatively small sample size of 515 individuals. Most studies utilised the Cerebral Performance Category (CPC) score to define neurological recovery, which is not in line with current consensus recommendations [63]. Future research with greater standardisation in reporting and assessment needs to be applied beyond survival

metrics if we are to improve functional recovery in the ICU-CA population.

We found that ICU mortality had an inverse relationship with the proportion of shockable rhythms, though this did not extend to hospital survival. A shockable rhythm has been associated with improved survival in non-ICU-CA [64–66]. Our results seem to suggest an early but transient survival advantage with a shockable rhythm, which are likely superseded by other factors once the patient is discharged from the ICU. For example, ICU-related complications adversely impact recovery and survival [67, 68]. Experiencing an ICU-CA may also result in updated goals-of-care discussions, further influencing treatment decisions and survival [69], though we were unable to further investigate these potential factors in the context of the current study. Given this, our results remain hypothesis-generating.

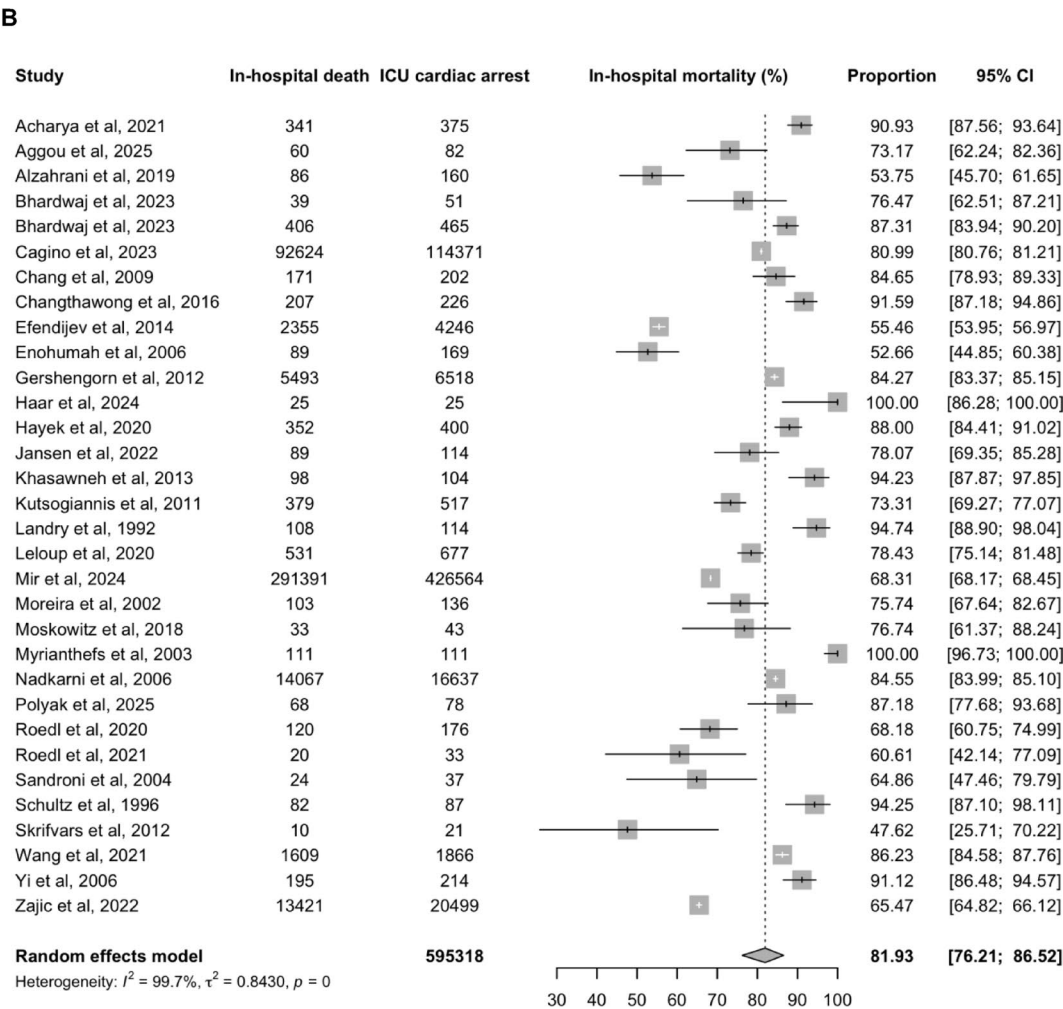
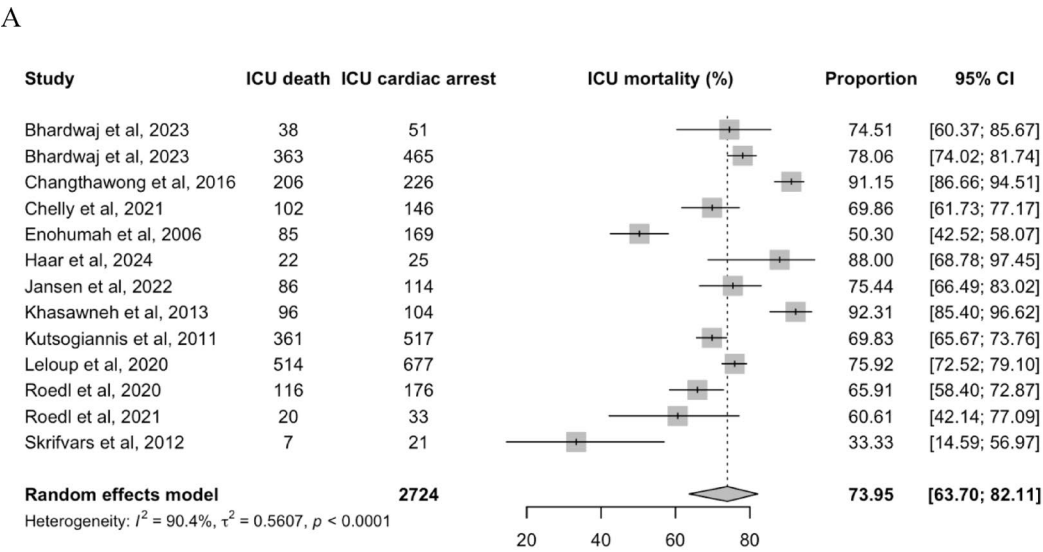


Fig. 4 A: Forest plot of ICU mortality in patients with ICU cardiac arrest. B: Forest plot of in-hospital mortality in patients with ICU cardiac arrest

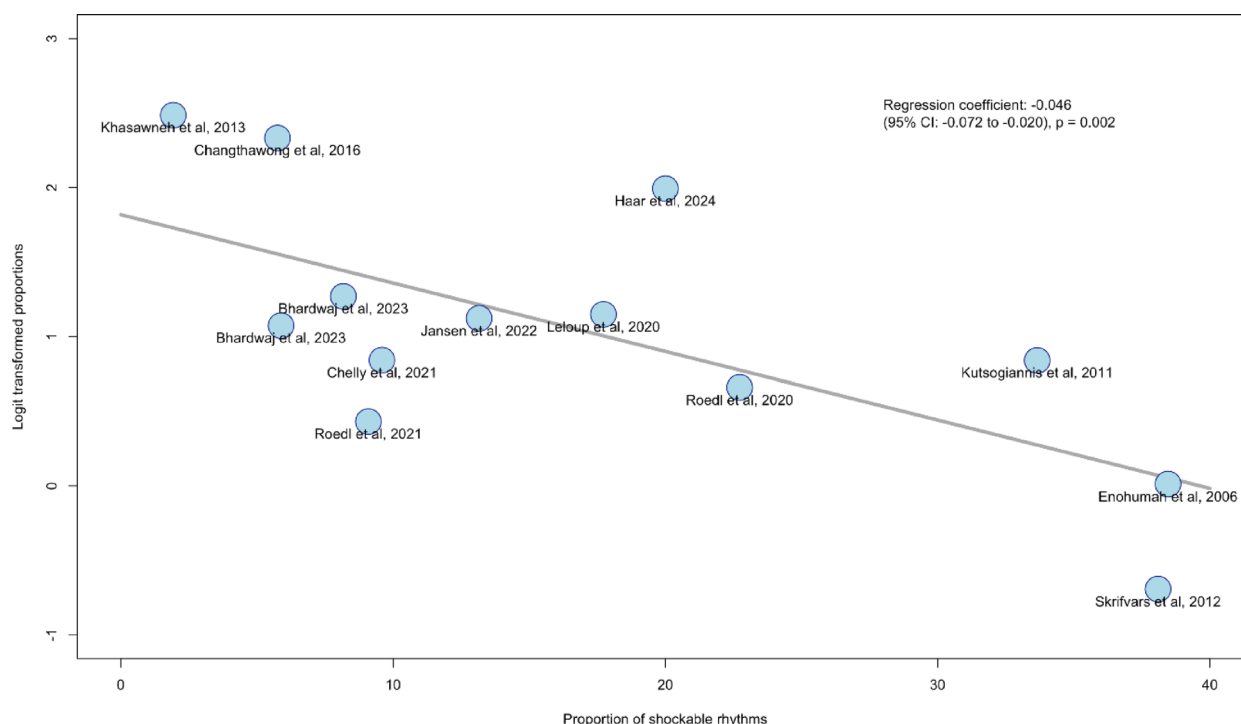


Fig. 5 Effect of proportion of shockable rhythms on ICU mortality

Limitations

We recognise several limitations of this study. We could not incorporate pertinent cardiac arrest characteristics into our analysis, including no-flow and low-flow times and post-arrest complications due to insufficient reporting among our included studies. Details on aetiology, contributing factors and level of support at time of arrest were only occasionally reported, and rarely in relation to outcomes. Although prespecified analyses of organ support modalities (e.g. vasoactive agents, mechanical ventilation, ECMO, and kidney replacement therapy) were planned, these were not performed because very few studies reported these outcomes, highlighting an important area for future research. We also opted not to focus on the role of extracorporeal CPR, given how sparsely it was utilised and reported amongst our included studies. Our search included patients managed in coronary care units, cardiac or cardiothoracic ICUs, a population that differs from the general ICU cohort in arrest aetiology (e.g. post-cardiotomy), access to specialised treatments (e.g. PCI), and cardiovascular comorbidity burden. We attempted to account for this heterogeneity through sensitivity analyses restricted to cardiac patients; however, this was feasible only for in-hospital mortality and was limited by small sample size. We also acknowledge international variability in the structure, monitoring intensity, and acuity of cardiac and cardiothoracic care units, which may have influenced observed outcomes. While meta-regression with publication year was performed,

we acknowledge that doing so may not fully capture the association of time with cardiac arrest outcomes as some studies stretched across long periods, and we did not have access to annual data for more granular analyses. Finally, our studies were clustered predominantly amongst high-income countries with robust ICU systems and workflows in place which may limit its generalisability to the rest of the world especially lower-income settings.

Conclusion

Cardiac arrest in the ICU remains associated with high mortality and morbidity. In COVID-19 individuals requiring ICU admission, the prevalence of cardiac arrest is higher compared to non-infected individuals. Prognostic factors in ICU-CA remain unclear. Further studies focusing on the ICU-CA population are warranted to improve prevention, survival and functional outcomes.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13054-026-05880-7>.

Supplementary Material 1

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Author contributions

Study conception: AS, DJ; Study design: OL, RRL, AS, DJ; Data collection: OL, VM; Data analysis and interpretation: OL, RRL, AS, DJ; Tables and figures: OL, VM; Writing of original draft: OL, RRL; All authors provided critical conceptual input, interpreted the data analysis, read, and approved the final draft. AS was responsible for the final decision to submit the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable. This study was prospectively registered with PROSPERO (CRD420251064966).

Consent for publication

Not applicable.

Competing interests

RRL acknowledges research funding support from the Clinician Scientist Development Unit, Yong Loo Lin School of Medicine, National University of Singapore. He is an editorial board member of *Critical Care*. SLL is supported by the National Medical Research Council Transitional Award; she has received research grants from National University Health System, National Kidney Foundation of Singapore and Singapore Heart Foundation. She is an Associate Editor of *Resuscitation*. All other authors do not have competing interests to declare.

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